

ECON 3350/7350 Applied Econometrics for Macroeconomics and Finance

Lecture 10: VAR Models (2)

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VAR Models

Recommended readings

Author	Title	Chapter	Call No
Enders	Applied Econometric Time Series, 4e	6	HB139 .E55 2015
Verbeek	A Guide to Modern Econometrics	9.5	HB139 .V465 2012

SVARs and Reduced Form VARs

Recall that combining two or more ARDL specifications leads to the **structural** VAR:

$$\mathbf{B}\mathbf{x}_t = \gamma_0 + \mathbf{\Gamma}_1\mathbf{x}_{t-1} + \boldsymbol{\varepsilon}_t,$$

$$E(\boldsymbol{\varepsilon}_t\boldsymbol{\varepsilon}_t') = \boldsymbol{\Sigma},$$

where $\mathbf{B}$ is invertible with ones on the diagonal and $\boldsymbol{\Sigma}$ is diagonal. The SVAR always has a corresponding **reduced form** VAR given by:

$$\mathbf{x}_t = \mathbf{B}^{-1}\gamma_0 + \mathbf{B}^{-1}\mathbf{\Gamma}_1\mathbf{x}_{t-1} + \mathbf{B}^{-1}\boldsymbol{\varepsilon}_t,$$

$$\mathbf{x}_t = \mathbf{a}_0 + \mathbf{A}_1\mathbf{x}_{t-1} + \mathbf{e}_t, \quad E(\mathbf{e}_t\mathbf{e}_t') = \boldsymbol{\Omega} \equiv \mathbf{B}^{-1}\boldsymbol{\Sigma}(\mathbf{B}^{-1})',$$

where $\boldsymbol{\Omega}$ is positive semi-definite, but not necessarily diagonal.

SVARs and Reduced Form VARs

In the **bivariate** case, the **structural shocks** ε_t are related to **reduced form errors** e_t by:

$$e_{1,t} = (\varepsilon_{1,t} - b_{12}\varepsilon_{2,t})/(1 - b_{12}b_{21}), \quad e_{2,t} = (\varepsilon_{2,t} - b_{21}\varepsilon_{1,t})/(1 - b_{12}b_{21}),$$

and the variances / covariances are related by:

$$\omega_1^2 = (\sigma_1^2 + b_{12}^2\sigma_2^2)/(1 - b_{12}b_{21})^2, \quad \omega_2^2 = (\sigma_2^2 + b_{21}^2\sigma_1^2)/(1 - b_{12}b_{21})^2,$$

$$\omega_{12} = -(b_{21}^2\sigma_1^2 + b_{12}^2\sigma_2^2)/(1 - b_{12}b_{21})^2.$$

VMA Representation of the VAR(1)

By repeated backwards substitution of the SVAR(1), we obtain

$$\begin{aligned}\mathbf{x}_t = & \left(\mathbf{I}_n + \mathbf{B}^{-1}\mathbf{\Gamma}_1 + \cdots + (\mathbf{B}^{-1}\mathbf{\Gamma}_1)^h \right) \mathbf{B}^{-1}\gamma_0 \\ & + \mathbf{B}^{-1}\varepsilon_t + \mathbf{B}^{-1}\mathbf{\Gamma}_1\mathbf{B}^{-1}\varepsilon_{t-1} + \cdots + (\mathbf{B}^{-1}\mathbf{\Gamma}_1)^h \mathbf{B}^{-1}\varepsilon_{t-h} \\ & + (\mathbf{B}^{-1}\mathbf{\Gamma}_1)^{h+1} \mathbf{x}_{t-h}.\end{aligned}$$

Recall that the VAR(1) is stationary as long as all eigenvalues of $\mathbf{A}_1 \equiv \mathbf{B}^{-1}\mathbf{\Gamma}_1$ are less than one in absolute value. In this case, $\mathbf{A}_1^h \equiv (\mathbf{B}^{-1}\mathbf{\Gamma}_1)^h \rightarrow 0$ as $h \rightarrow \infty$ and we can write

$$\mathbf{x}_t = (\mathbf{I}_n - \mathbf{A}_1)^{-1} \mathbf{a}_0 + \sum_{h=0}^{\infty} \mathbf{A}_1^h \mathbf{B}^{-1} \varepsilon_{t-h}.$$

This is the VMA(∞) representation of the VAR(1).

Impulse Response Functions

The $n \times n$ matrix $\Phi_h = \mathbf{A}_1^h \mathbf{B}^{-1}$ contains all the **impulse responses** at horizon h .

Specifically, the (i, j) th element of Φ_h , denoted $\phi_{h,ij}$, is the impulse response of variable i to shock j at h periods after impact.

The **impulse response function** (IRF), defined for variable i to a given shock j , plots impulse responses against the time horizon $0, 1, \dots, h$ periods after impact.

In the bivariate case, we might obtain the IRF for x_1 to shocks in ε_2 from

$$\begin{pmatrix} x_{1,t} \\ x_{2,t} \end{pmatrix} = \begin{pmatrix} \bar{x}_1 \\ \bar{x}_2 \end{pmatrix} + \begin{pmatrix} \phi_{0,11} & \phi_{0,12} \\ \phi_{0,21} & \phi_{0,22} \end{pmatrix} \begin{pmatrix} \varepsilon_{1,t} \\ \varepsilon_{2,t} \end{pmatrix} \\ + \begin{pmatrix} \phi_{1,11} & \phi_{1,12} \\ \phi_{1,21} & \phi_{1,22} \end{pmatrix} \begin{pmatrix} \varepsilon_{1,t-1} \\ \varepsilon_{2,t-1} \end{pmatrix} + \begin{pmatrix} \phi_{2,11} & \phi_{2,12} \\ \phi_{2,21} & \phi_{2,22} \end{pmatrix} \begin{pmatrix} \varepsilon_{1,t-2} \\ \varepsilon_{2,t-2} \end{pmatrix} + \dots$$

Forecast Error Variance Decompositions

The variance of the h -step ahead forecast error for variable i is

$$\begin{aligned}\text{Var}(x_{i,t+h} - \bar{x}_i) &= \sum_{j=1}^n \phi_{0,ij}^2 \sigma_j^2 + \sum_{j=1}^n \phi_{1,ij}^2 \sigma_j^2 + \cdots + \sum_{j=1}^n \phi_{h,ij}^2 \sigma_j^2, \\ &= \underbrace{\sum_{s=0}^h \phi_{s,i1}^2 \sigma_1^2}_{\text{contrib of } \varepsilon_1} + \underbrace{\sum_{s=0}^h \phi_{s,i2}^2 \sigma_2^2}_{\text{contrib of } \varepsilon_2} + \cdots + \underbrace{\sum_{s=0}^h \phi_{s,in}^2 \sigma_n^2}_{\text{contrib of } \varepsilon_n}.\end{aligned}$$

We can use this to evaluate how much of the **forecast error variance** of variable i at horizon h is explained by shock j , **relative to all other shocks**:

$$\text{FEV}_{h,ij} = \frac{\sum_{s=0}^h \phi_{s,ij}^2 \sigma_j^2}{\sum_{s=0}^h \phi_{s,i1}^2 \sigma_1^2 + \sum_{s=0}^h \phi_{s,i2}^2 \sigma_2^2 + \cdots + \sum_{s=0}^h \phi_{s,in}^2 \sigma_n^2}.$$

Identification of IRFs and FEVDs

Both IRFs and FEVDs measure the **dynamic effect** of a structural shock ε_j on a variable x_i .

Both can be computed easily from the VMA(∞) representation, written in lag-polynomial form as

$$\mathbf{x}_t = \bar{\mathbf{x}} + \Phi(L)\varepsilon_t,$$

where $\Phi(L) = \Phi_0 + \Phi_1 L + \dots$ and $\Phi_h = \mathbf{A}_1^h \mathbf{B}^{-1}$.

Note that this formulation is valid for VARs with **any lag length** p —just rewrite a VAR(p) as a VAR(1) using the **companion form**.

From data, we can easily estimate $\mathbf{A}_1$ using the reduced form VAR (and, e.g., least squares methods). The trouble is with estimating $\mathbf{B}$.

Identification of IRFs and FEVDs

Two ways to estimate $\mathbf{B}$:

1. estimate the SVAR directly (OLS may no longer work, but can use more complex methods such as MLE);
2. estimate the reduced form VAR, then compute $\Sigma, \mathbf{B}, \Gamma_1$ from estimates of $\Omega, \mathbf{A}_1$.

Either approach requires **identifying restrictions**. We will focus on the second.

As previously discussed, the key issue is that there are many SVARs that correspond to any given VAR.

- This means that times series data alone is **not enough** to identify all the parameters of an SVAR; we need additional information.

Identification Using Exclusion Restrictions

The common approach we will consider is to compute:

1. $\mathbf{B}^{-1}\boldsymbol{\Sigma}(\mathbf{B}^{-1})' = \boldsymbol{\Omega}$,
2. $\boldsymbol{\Gamma}_1 = \mathbf{B}\mathbf{A}_1$.

To implement this, we need to impose restrictions on $\mathbf{B}$ and $\boldsymbol{\Sigma}$ such that the decomposition $\mathbf{B}^{-1}\boldsymbol{\Sigma}(\mathbf{B}^{-1})' = \boldsymbol{\Omega}$ is **unique**.

Many approaches have been developed to achieve this in various contexts. The most common of these amount to imposing **zero restrictions** (or **exclusion restrictions**) on either the elements of $\mathbf{B}$ or the elements of $\mathbf{C} = \mathbf{B}^{-1}$.

The basic idea is that **economic theory** often elicits strong implications for the relationships between reduced form errors and structural shocks, i.e.

$$e_t = \mathbf{C}\boldsymbol{\varepsilon}_t.$$

We will discuss two types of factorizations:

1. **Cholesky factorization**: exact identification obtained by setting $\Sigma = \mathbf{I}_n$ and $\mathbf{C}$ to a lower (sometimes upper) triangular matrix; this generates a **recursive** structure for the set of equations $e_t = \mathbf{C}\varepsilon_t$;
2. **structural factorization**: possibly **non-recursive** and/or **over-identified** structures for $e_t = \mathbf{C}\varepsilon_t$ obtained by setting elements of $\mathbf{C}$ to zero according to theoretical considerations.

Example of a Cholesky factorisation:

$$\begin{pmatrix} e_{1,t} \\ e_{2,t} \\ e_{3,t} \\ e_{4,t} \\ e_{5,t} \end{pmatrix} = \begin{pmatrix} c_{11} & 0 & 0 & 0 & 0 \\ c_{21} & c_{22} & 0 & 0 & 0 \\ c_{31} & c_{32} & c_{33} & 0 & 0 \\ c_{41} & c_{42} & c_{43} & c_{44} & 0 \\ c_{51} & c_{52} & c_{53} & c_{54} & c_{55} \end{pmatrix} \begin{pmatrix} \varepsilon_{1,t} \\ \varepsilon_{2,t} \\ \varepsilon_{3,t} \\ \varepsilon_{4,t} \\ \varepsilon_{5,t} \end{pmatrix}.$$

SVAR Example with Structural Factorization

Consider estimating a VAR with

$$\mathbf{x}_t = (\text{HKRET}_t, \text{SINGRET}_t, \text{KOREARET}_t, \text{JAPRET}_t, \text{THAIRET}_t)'$$

Identification is achieved with the structural factorization:

$$\begin{pmatrix} e_{1,t} \\ e_{2,t} \\ e_{3,t} \\ e_{4,t} \\ e_{5,t} \end{pmatrix} = \begin{pmatrix} c_{11} & c_{12} & 0 & 0 & 0 \\ 0 & c_{22} & 0 & 0 & 0 \\ c_{31} & c_{32} & c_{33} & c_{34} & 0 \\ c_{41} & c_{42} & 0 & c_{44} & 0 \\ c_{51} & c_{52} & c_{53} & c_{54} & c_{55} \end{pmatrix} \begin{pmatrix} \varepsilon_{1,t} \\ \varepsilon_{2,t} \\ \varepsilon_{3,t} \\ \varepsilon_{4,t} \\ \varepsilon_{5,t} \end{pmatrix}.$$

Useful Properties of Matrix Inverses

The following identities are often useful in working with identifying restrictions:

$$\begin{pmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{C} & \mathbf{D} \end{pmatrix}^{-1} = \begin{pmatrix} \mathbf{A}^{-1} & \mathbf{0} \\ -\mathbf{D}^{-1}\mathbf{C}\mathbf{A}^{-1} & \mathbf{D}^{-1} \end{pmatrix}, \quad \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{0} & \mathbf{D} \end{pmatrix}^{-1} = \begin{pmatrix} \mathbf{A}^{-1} & -\mathbf{A}^{-1}\mathbf{B}\mathbf{D}^{-1} \\ \mathbf{0} & \mathbf{D}^{-1} \end{pmatrix}.$$

We use identities such as these to check which zero restrictions on $\mathbf{C}$ correspond to zero restrictions on $\mathbf{B}$.

- If $\mathbf{C}$ is *lower-triangular*, then $\mathbf{B}$ is also *lower triangular*.

In terms of the current SVAR example,

$$\begin{pmatrix} c_{11} & c_{12} & 0 & 0 & 0 \\ 0 & c_{22} & 0 & 0 & 0 \\ c_{31} & c_{32} & c_{33} & c_{34} & 0 \\ c_{41} & c_{42} & 0 & c_{44} & 0 \\ c_{51} & c_{52} & c_{53} & c_{54} & c_{55} \end{pmatrix}^{-1} = \begin{pmatrix} b_{11} & b_{12} & 0 & 0 & 0 \\ 0 & b_{22} & 0 & 0 & 0 \\ b_{31} & b_{32} & b_{33} & b_{34} & 0 \\ b_{41} & b_{42} & 0 & b_{44} & 0 \\ b_{51} & b_{52} & b_{53} & b_{54} & b_{55} \end{pmatrix}$$

SVAR Estimates

Structural VAR Estimates
 Date: 05/01/13 Time: 23:16
 Sample (adjusted): 1/08/1985 4/28/1999
 Included observations: 3732 after adjustments
 Estimation method: method of scoring (analytic derivatives)
 Convergence achieved after 7 iterations
 Structural VAR is just-identified

Model: $Ae = Bu$ where $E[uu'] = I$
 Restriction Type: short-run text form
 $@e1 = c(1)*@e2 + c(2)*@u1$
 $@e2 = c(3)*@u2$
 $@e3 = c(4)*@e1 + c(5)*@e2 + c(6)*@u3 + c(7)*@e4$
 $@e4 = c(8)*@e1 + c(9)*@e2 + c(10)*@u4$
 $@e5 = c(11)*@e1 + c(12)*@e2 + c(13)*@e3 + c(14)*@e4 + c(15)*@u5$
 where
 @e1 represents HKRET residuals
 @e2 represents SINGRET residuals
 @e3 represents KOREARET residuals
 @e4 represents JAPANRET residuals
 @e5 represents THAIARET residuals

	Coefficient	Std. Error	z-Statistic	Prob.
C(1)	0.573645	0.017930	31.99279	0.0000
C(4)	0.052283	0.017272	3.027013	0.0025
C(5)	0.089572	0.021865	4.096624	0.0000
C(7)	0.039471	0.020956	1.883512	0.0596
C(8)	0.099218	0.013393	7.407963	0.0000
C(9)	0.255050	0.016561	15.40072	0.0000
C(11)	0.102636	0.016945	6.057076	0.0000
C(12)	0.200562	0.021472	9.340506	0.0000
C(13)	0.073377	0.016039	4.574804	0.0000
C(14)	0.017890	0.020544	0.870832	0.3838
C(2)	0.015372	0.000178	86.39444	0.0000
C(3)	0.014034	0.000162	86.39444	0.0000
C(6)	0.016102	0.000186	86.39444	0.0000
C(10)	0.012578	0.000146	86.39444	0.0000
C(15)	0.015778	0.000183	86.39444	0.0000

Log likelihood 52250.15

SVAR Estimates

Estimated A matrix:

1.000000	-0.573645	0.000000	0.000000	0.000000
0.000000	1.000000	0.000000	0.000000	0.000000
-0.052283	-0.089572	1.000000	-0.039471	0.000000
-0.099218	-0.255050	0.000000	1.000000	0.000000
-0.102636	-0.200562	-0.073377	-0.017890	1.000000

Estimated B matrix:

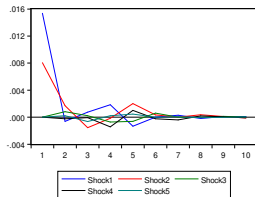
0.015372	0.000000	0.000000	0.000000	0.000000
0.000000	0.014034	0.000000	0.000000	0.000000
0.000000	0.000000	0.016102	0.000000	0.000000
0.000000	0.000000	0.000000	0.012578	0.000000
0.000000	0.000000	0.000000	0.000000	0.015778

SVAR Estimates

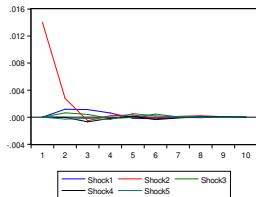
Variance Decomposition of SINGRET:						
Period	S.E.	Shock1	Shock2	Shock3	Shock4	Shock5
1	0.014034	0.000000	100.0000	0.000000	0.000000	0.000000
2	0.014377	0.677737	99.07566	0.204834	6.38E-05	0.041701
3	0.014454	1.276482	98.15505	0.285502	0.222962	0.060004
4	0.014476	1.464289	97.87940	0.307290	0.233731	0.115293
5	0.014491	1.474237	97.72057	0.307005	0.244784	0.253401
6	0.014507	1.496842	97.52127	0.406663	0.303717	0.271507
7	0.014508	1.497325	97.51182	0.407598	0.311449	0.271807
8	0.014510	1.500044	97.50597	0.407660	0.314421	0.271901
9	0.014510	1.500185	97.50386	0.409339	0.314684	0.271935
10	0.014511	1.502845	97.50074	0.409362	0.315029	0.272029

SVAR IRFs

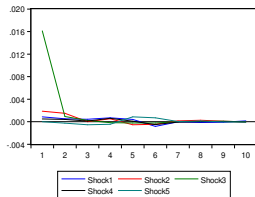
Response of HKRET to Structural
One S.D. Innovations



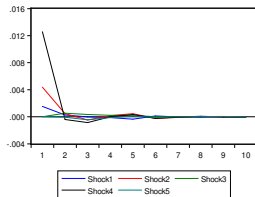
Response of SINGRET to Structural
One S.D. Innovations



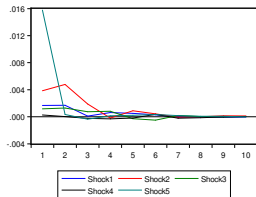
Response of KOREARET to Structural
One S.D. Innovations



Response of JAPANRET to Structural
One S.D. Innovations



Response of THAIRET to Structural
One S.D. Innovations



Granger (Non-)Causality

Granger causality is a statement about whether **past information** on some variable, say z_t , improves the forecast of another variable, say y_t , where both z_t and y_t are endogenous variables in a VAR.

Formally, if

$$E(y_{t+1} | z_t, \dots, z_0, y_t, \dots, y_0) = E(y_{t+1} | y_t, \dots, y_0),$$

then z_t **does not Granger-cause** y_t .

Unlike IRFs and FEVs, Granger (non-)causality is derived entirely from the reduced form VAR (so no additional restrictions are necessary).

Two things to beware of:

1. data-based tests can confirm Granger **causality** but they **cannot confirm Granger non-causality**;
2. Granger non-causality **does not** imply **exogeneity** because y_t can still be **contemporaneously** correlated with z_t .

Testing for Granger Non-Causality

If the VAR

$$\mathbf{x}_t = \mathbf{a}_0 + \mathbf{A}_1 \mathbf{x}_{t-1} + \cdots + \mathbf{A}_p \mathbf{x}_{t-p} + e_t$$

is **stable**, then we only need to do an F -test.

To test that $x_{j,t}$ does not Granger-cause $x_{i,t}$, specify

$$H_0 : a_{1,ij} = a_{2,ij} = \cdots = a_{p,ij},$$

where $a_{l,ij}$ is the (i, j) th element of $\mathbf{A}_l$, then implement a standard F -test on the restrictions.

For the non-stationary case, an MWALD test can be used instead.